In [ ]:
```
EXPERIMENT 3 – CREATING DATAFRAMES & ANALYZING PROPERTIES
PIN NO: 241UDS0109

AIM:
To create Pandas DataFrames using NumPy arrays, CSV/Excel files, and analyze DataFrame properties using structured
data.
```

In [ ]:
```
# Step 1: Install and Import Libraries
If NumPy or Pandas is not installed, run %pip install numpy pandas in a separate cell.
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
# Step 2: Create a 2-D Array
Rows represent library branches; columns represent monthly visitor counts.
```

In [2]:
```
library_visitors_241UDS0109 = np.array([
    [1180, 1325, 1460],
    [940, 1085, 1160],
    [790, 880, 965]
])
library_visitors_241UDS0109
```

Out[2]:
```
array([[1180, 1325, 1460],
       [ 940, 1085, 1160],
       [ 790,  880,  965]])
```

In [3]:
```
library_visitors_241UDS0109.shape, library_visitors_241UDS0109.ndim
```

Out[3]:
```
((3, 3), 2)
```

In [ ]:
```
# Step 3: Convert the 2-D Array into a DataFrame
Set branch names as the index and months as the columns.
```

In [4]:
```
library_index_241UDS0109 = ["Central", "North", "South"]
library_columns_241UDS0109 = ["January", "February", "March"]
library_df_241UDS0109 = pd.DataFrame(
    data=library_visitors_241UDS0109,
    index=library_index_241UDS0109,
    columns=library_columns_241UDS0109
)
library_df_241UDS0109
```

Out[4]:

	January	February	March
Central	1180	1325	1460
North	940	1085	1160
South	790	880	965

In [ ]:
```
## Add a new column
Add April visitor counts. The number of values must match the number of branches.
```

In [5]:
```
library_df_241UDS0109["April"] = [1510, 1215, 1010]
library_df_241UDS0109
```

Out[5]:

	January	February	March	April
Central	1180	1325	1460	1510
North	940	1085	1160	1215
South	790	880	965	1010

In [ ]:
```
### Add a new branch (row)
Add an East branch with values for each month.
```

In [6]:
```
new_branch_241UDS0109 = pd.DataFrame(
    {"January": [735], "February": [810], "March": [875], "April": [925]},
    index=["East"]
)
library_df_241UDS0109 = pd.concat([library_df_241UDS0109, new_branch_241UDS0109])
library_df_241UDS0109
```

Out[6]:

	January	February	March	April
Central	1180	1325	1460	1510
North	940	1085	1160	1215
South	790	880	965	1010
East	735	810	875	925

In [ ]:
```
# Step 4: Creating a DataFrame with Existing CSV/Excel File
Read the new CSV dataset provided with this notebook.
```

In [7]:
```
library_data_241UDS0109 = pd.read_csv("Experiment_3_241UDS0109_Library_Visitors.csv")
library_data_241UDS0109
```

Out[7]:

	PIN	Branch	Month	Visitors	Books_Issued	Study_Hours	Membership_Type
0	241UDS0109	Central	January	1180	720	395	Student
1	241UDS0109	Central	February	1325	805	440	Faculty
2	241UDS0109	Central	March	1460	875	485	Student
3	241UDS0109	North	January	940	585	330	Student
4	241UDS0109	North	February	1085	655	365	Faculty
5	241UDS0109	North	March	1160	710	400	Student
6	241UDS0109	South	January	790	485	275	Student
7	241UDS0109	South	February	880	535	305	Faculty
8	241UDS0109	South	March	965	590	335	Student

In [ ]:
```
## Step 5: Analyze the Important Properties of the DataFrame
Use `head()`, `shape`, `index`, `dtypes`, and `columns` to inspect the data.
```

In [8]:
```
library_data_241UDS0109.head()
```

Out[8]:

	PIN	Branch	Month	Visitors	Books_Issued	Study_Hours	Membership_Type
0	241UDS0109	Central	January	1180	720	395	Student
1	241UDS0109	Central	February	1325	805	440	Faculty
2	241UDS0109	Central	March	1460	875	485	Student
3	241UDS0109	North	January	940	585	330	Student
4	241UDS0109	North	February	1085	655	365	Faculty

In [9]:
```
library_data_241UDS0109.shape
```

Out[9]:
```
(9, 7)
```

In [10]:
```
library_data_241UDS0109.index
```

Out[10]:
```
RangeIndex(start=0, stop=9, step=1)
```

In [11]:
```
library_data_241UDS0109.dtypes
```

Out[11]:
```
PIN              object
Branch           object
Month            object
Visitors          int64
Books_Issued      int64
Study_Hours       int64
Membership_Type  object
dtype: object
```

In [12]:
```
library_data_241UDS0109.columns
```

Out[12]:
```
Index(['PIN', 'Branch', 'Month', 'Visitors', 'Books_Issued', 'Study_Hours',
       'Membership_Type'],
      dtype='object')
```

In [ ]:
```
## Step 6: Information About the DataFrame
`info()` reports row count, non-null values, data types, and memory usage.
```

In [13]:
```
library_data_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 7 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   PIN              9 non-null      object
 1   Branch           9 non-null      object
 2   Month            9 non-null      object
 3   Visitors         9 non-null      int64 
 4   Books_Issued     9 non-null      int64 
 5   Study_Hours      9 non-null      int64 
 6   Membership_Type  9 non-null      object
dtypes: int64(3), object(4)
memory usage: 636.0+ bytes
```

In [ ]:
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## Step 7: Descriptive Statistics
Use `describe()` to summarize the numeric columns.
```

In [14]:
```
library_data_241UDS0109.describe()
```

Out[14]:

	Visitors	Books_Issued	Study_Hours
count	9.000000	9.000000	9.00000
mean	1087.222222	662.222222	370.00000
std	217.170888	127.527230	66.84871
min	790.000000	485.000000	275.00000
25%	940.000000	585.000000	330.00000
50%	1085.000000	655.000000	365.00000
75%	1180.000000	720.000000	400.00000
max	1460.000000	875.000000	485.00000

In [ ]:
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## Step 8: Transpose Rows and Columns
Display the first 10 rows and transpose them. Since there are 9 rows, all records are shown.
```

In [15]:
```
library_data_241UDS0109.head(10).transpose()
```

Out[15]: Columns 1–8 of 10

	0	1	2	3	4	5	6
PIN	241UDS0109	241UDS0109	241UDS0109	241UDS0109	241UDS0109	241UDS0109	241UDS0109
Branch	Central	Central	Central	North	North	North	South
Month	January	February	March	January	February	March	January
Visitors	1180	1325	1460	940	1085	1160	790
Books_Issued	720	805	875	585	655	710	485
Study_Hours	395	440	485	330	365	400	275
Membership_Type	Student	Faculty	Student	Student	Faculty	Student	Student

Columns 9–10 of 10 · first column repeated

	7	8
PIN	241UDS0109	241UDS0109
Branch	South	South
Month	February	March
Visitors	880	965
Books_Issued	535	590
Study_Hours	305	335
Membership_Type	Faculty	Student

In [ ]:
```
## RESULT
Using NumPy and Pandas, a DataFrame was created from a 2-D array, a column and row were added, a CSV file was
loaded, and DataFrame properties, information, descriptive statistics, and transpose were examined.
```